

CH4203: Organic Synthesis Laboratory

Expt. No. 7: Organocatalytic Asymmetric Michael Reaction of Cyclohexanone with β -Nitrostyrene

Home Assignment

Due date: 7 days after the expt. done

(Q1) Write down the mechanism of Organocatalytic Michael Reaction of Cyclohexanone with β -Nitrostyrene.

[1+2+2 = 5]

(Q2) Why the diastereoselectivity observed in >20:1 ratio? Explain this by proposing a transition state clearly showing how enamine is helping in induction of enantioselectivity and diastereoselectivity.

[1+2+2 = 5]